Chaithanya Naik

Member of Technical Staff
R&D - Avi Networks
VMware Inc.

chaithanyampcrl@gmail.com
metalcyanide
chaithanya-naik

Research Interests

Quantum Computing and Information, Quantum Control, Quantum algorithms, Machine learning and AI

Education

Indian Institute of technology Bombay 2020-2021
M.Tech in Computer Science and Engineering **GPA (Thesis) : 10.0/10.0**
Advisor: Prof. Sai Vinjanampathy

Indian Institute of technology Bombay 2016-2020
B.Tech in Computer Science and Engineering **GPA: 7.25/10.0**
Minor in Physics
Minor in AI and Data Science
Advisors: Prof. Amit Sethi, Prof. Ashutosh Gupta

Research Experience

Robust Quantum Optimal Control *Spring 2021*
Guide: Prof. Sai Vinjanampathy *Master's Thesis Project, IIT Bombay*
Submitted abstract to APS March meeting 2022

- Designed sequence-to-sequence learning model using deep networks based on encoder-decoder architecture consisting of CRNNs to generate the optimized control sequences for transmon and cross resonance systems despite noise
- Enhanced the performance of quantum gate compilations by 100x compared to current state of the art solutions
- Analyzed literature on Geodesic analysis, Deep Sequential models, and Optimal Quantum Control methods

Controller Synthesis *Spring 2021*
Guide: Prof. Ashutosh Gupta *R&D Project, IIT Bombay*

- Synthesised a controller for real-time system, based on data-driven RL and algorithmic SMT/SAT approaches to control a railway network modelled as a network of timed automata constrained over a set of specifications
- Utilised tools like UPPAAL and DCValid to design and model networks of timed automata and verify solutions
- Studied about determinization and minimization of timed automata for a specification given in duration calculus

Modular Quantum Computer Design *Autumn 2020*
Guide: Prof. Sai Vinjanampathy *Master's Thesis Project, IIT Bombay*

- Designed robust control protocols and generalized for n-Qubit Gates to enable fault-tolerant quantum computation
- Implemented a modular RL framework for determining the optimal pulse sequence for any general quantum gate preparation using Deep RL through DQN, DDPG, Trust Region Policy Optimization, and Actor-Critic methods

Interactive Image Segmentation *Autumn 2020*
Guide: Prof. Amit Sethi *Bachelor's Thesis Project, IIT Bombay*

- Automated the generation of markers required to segment desired object in image with minimal user intervention
- Designed a deep network using prior based feature enrichment network (PFE-net) and feature back-propagating refinement scheme (f-BRS) on PASCAL-5i dataset, to decrease the number of user clicks required
- Experimented with SeedNet, a DQN approach over MSRA10k dataset to achieve IOU of around 60.5

Quantum Gate Optimization *Autumn 2019*
Guide: Prof. Sai Vinjanampathy *R&D Project, IIT Bombay*

- Implemented Subspace Selective Self-Adaptive Differential Evolution (SUSSADE) technique to generate the instructions for effectively determining the control parameters of the quantum control scheme
- Critically analyzed advantages of GrAPE with Push-Pull optimization and RL approaches like Q-learning, Deep-Deterministic Policy gradient methods and evolutionary RL for improving existing bounds for gate fidelity

Professional Experience

VMware Inc. *July 2021 - present*
Avi NSX-T load-balancer | Mentor: Anurag Palsule *Bangalore, India*

- Working on enhancing the performance of load balancer to manage heavy workloads through kubernetes operator
- Part of the team maintaining an open source project, catering to needs for NSX-T and openshift platforms
- Worked on functional test-suites to ensure better performance for individual components

Amazon Development Centre India Pvt. Ltd *May - July 2019*
Auto Trouble Ticket Manager | Mentor: Archit Agrawal *Hyderabad, India*

- Automated process of resolving trouble tickets raised on amazon services by developing java package
- Designed runtime compilation module using Java Compiler API to implement dynamic code execution by converting java snippet code from different source packages into a compiled java object
- Developed a generic framework to plug-in their models for information retrieval and auto-suggesting resolution

OYO - Oravel Stays Pvt. Ltd. *May - July 2018*
Price Elasticity estimation | Mentor: Rahul Gupta *Gurugram, India*

- Designed a **prediction model** in **R** that predicts the likelihood of achieving certain profit from the current situation and gives the optimized path of **price movement** using probability-based analysis
- Implemented density-based clustering using **HDBSCAN density cluster model** to remove the outliers
- Implemented **two-stage least squares** instrumental variable model to remove biases in explanatory variables

Academic Projects

Gomoku RL Playing Agent | Fundamentals of Intelligent agents *Autumn 2020*

- Created Gomuku playing AI agents using Monte Carlo Tree Search (MCTS) algorithm in C++
- Implemented efficient guided search using various heuristics to create different learning AI agents using reward shaping, setting up intelligent priors and utilized multi-threading to carryout rollouts in parallel

Depth Map Prediction From Single Image | Computer Vision *Spring 2019*

- Built two-stack CNN-Residual Network model to estimate depth map from a single RGB image
- Implemented transfer learning using pre-trained ResNet-50 network to enhance the performance

Compiler for C-like language | Implementation of Programming Languages *Spring 2019*

- Developed a compiler and evaluator for a subset of C supporting functions, scope levels, and control sequences
- Utilized Lex for tokenizing, Yacc for parsing and constructed AST to generate MIPS assembly code

Features of XV6 | Operating Systems *Autumn 2018*

- Examined xv6 source code and implemented process scheduling algorithms like round robin and priority-based.
- Implemented Memory management techniques like lazy page allocation and applications of pthreads

Fake News Detection by Crowdsourcing | Database and Information Systems *Autumn 2018*

- Developed web and android App for crowdsourcing the verification of spurious news articles.
- Designed a database schema for users, volunteers and admins providing tools to review, appoint and approve
- Implemented routing algorithms to distribute tasks among volunteers based on domain-specific knowledge

Microarchitectural Attacks | Computer Architecture *Autumn 2018*

- Implemented FLUSH+RELOAD attack to extract the private key from the GnuPG implementation of RSA
- Implemented Cache Template Attack to profile and exploit cache-based information leakage of programs
- Proposed automated DRAMA Template attack by reverse engineering DRAM addressing and template attack

3D Modelling and Animation | Computer Graphics *Autumn 2018*

- Designed 3D graphical models through hierarchical modeling in C++ OpenGL with textures, shading, and lighting
- Implemented framework to create dynamic Bezier curves through clicked control points for camera motion

Socializing - Social Networking Platform | Software Systems Lab *Autumn 2017*

- Developed Django based web application which serves as a social platform to interact through posts and messages
- Implemented real-time chatbox using Django channels with the help of websockets

Awards & Achievements

- Secured 1st position in ISRO's web visualization tool for Astrosat data at 9th Inter IIT Tech Meet *2021*
- Bagged Gold Medal in Bosch's Route Optimization challenge at the 8th Inter IIT Tech Meet *2019*
- Secured 2nd position in Capture The Flag cybersecurity challenge at the 8th Inter IIT Tech Meet *2019*
- Bagged Gold Medal at the 7th Inter IIT Tech Meet in Star Cluster Identifier competition *2018*
- Secured AIR 340 in IIT-JEE Advanced | AIR 176 in JEE Main Paper-I | AIR 90 in JEE Main Paper-II *2016*
- Amongst Top 300 students in the country qualified for INPhO (for two straight years) and INMO *2015,14*
- Achieved KVPY Fellowship with All India Rank 173 organized under DST, Government of India *2015*
- Recipient of NTSE Scholarship awarded by NCERT under Ministry of HRD, Government of India *2012*

Teaching & Mentoring Experience

Teaching Assistant *July 2020 - June 2021*

- Responsible for clarifying queries of students, helping with course material and grading assignments
- Courses Taught - Quantum Computing and Information (Spring 2021) and python programming lab (Autumn 2020)

Mentor
Quantum Computing, Computer Vision: Maths and Physics Club *April - June 2020*

- Spoof-Resistant Facial Recognition using Deep Learning - guided a group of 4 students
- Designing Rubik's Cube Solver using Reinforcement Learning using DeepCubeA - guided a group of 6 students
- Project on creating star hopping guide & creating astronomy website using Django - guided a group of 6 students
- Helped a total of 12 students in pursuing their interests in Quantum Computing, Astronomy & Computer Vision

Department Academic Mentor | CSE Department, IIT Bombay *June 2020 - July 2021*

- Mentoring 4 sophomores for their academic and general concerns, and helping them cope with the curriculum
- Mentor to additional 2 students in an academic rehabilitation program (ARP) & guiding them back on track

Positions of Responsibility

Department General Secretary | CSE Department, IIT Bombay *April 2019 - July 2020*

- Elected student representative in department policy formulation committees to ensure student involvement
- Re-instantiated and managed Cyber-Security Club, organized CTF events to increase awareness about cybersecurity
- Spearheaded a 3-tier team of 12 members in designing and development of the content for articles pertaining to research groups, the scope of career in core and non-core sectors and semester exchange experience

Convener | Krittika - The Astronomy Club *April 2017 - March 2018*

- Organized numerous events like series of lectures, trips to GMRT and Nehru planetarium, star gazing sessions.
- Efficiently managed a budget of **200 thousand INR** and improved the outreach of club in the institute

Technical Skills

Programming	C/C++, Python, Java, R, Racket(Scheme), SWI-Prolog, Bash, VHDL
Deep Learning	PyTorch, TensorFlow, Keras, TensorFlow Quantum

Extracurriculars

- Part of the Inter IIT contingent securing Runner's Up Position at Inter IIT Tech Meet held at IIT Bombay *2018*
- Awarded Hostel Player of GC (Carrom) for best performance in Inter Hostel Sports, IIT Bombay *2016*
- Completed a year-long course in Athletics offered by National Sports Organization (NSO), IIT Bombay *2016*
- Attended Vijyoshi Science Camp organized by Indian Institute of Science (IISc), Bengaluru, India *2015*